Average Rate of Change

Computing average rate of change

1 Find the average rate of change of $f(x) = x^2$ over each interval:

a $[0, 2]$ **b** $[1, 3]$ **c** $[2, 5]$ **d** $[-1, 1]$

2 Find the average rate of change of $f(x) = 2^x$ over the interval $[0, 3]$.

3 Show that the average rate of change of $g(x) = 3x - 5$ is the same over any interval $[a, b]$, and state its value.

Applications

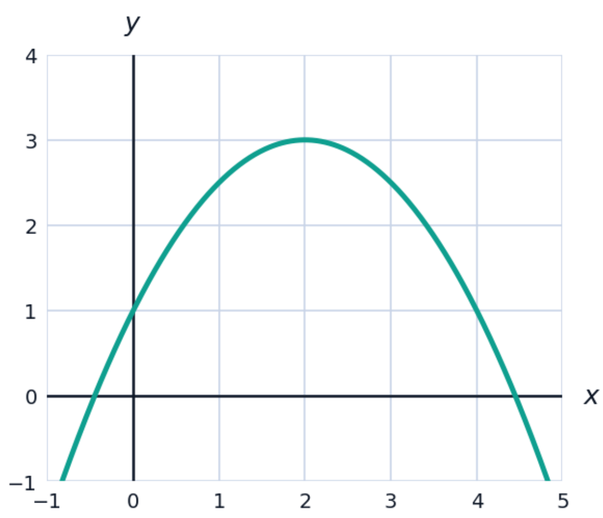
4 A ball is thrown upward so that its height after t seconds is $h(t) = -5t^2 + 20t$ meters.

a Find the average rate of change of the height over $[0, 2]$.

b Find the average rate of change of the height over $[1, 3]$.

c Interpret your answer to part b in the context of the ball's motion.

5 The graph of $f(x) = -\frac{1}{2}x^2 + 2x + 1$ is shown. Estimate the average rate of change over $[0, 4]$, then verify your estimate algebraically.



6 A company's revenue (in thousands of riyals) is modeled by $R(t) = 40 + 6t - 0.5t^2$, where t is years since 2020.

a Find the average rate of change of revenue over $[0, 4]$.

b Find the average rate of change of revenue over $[4, 8]$.

c Compare your two answers and interpret what they say about the company's growth.

Comparing rates and estimating from a table

7 A table of values for $g(x)$ is given: $g(1) = 5$, $g(3) = 17$, $g(6) = 44$, $g(10) = 92$. Find the average rate of change of g over:

a $[1, 3]$

b $[3, 6]$

c $[6, 10]$

8 Based on the previous question, is the average rate of change of g increasing, decreasing, or constant as x increases? What does this suggest about the shape of g's graph?

9 Two functions model distance traveled: $d_1(t) = 5t$ and $d_2(t) = t^2$ (in km, t in hours). Find the average speed (average rate of change) of each over $[2, 6]$, and state which traveler was faster on average.

Average rate of change and secant lines

10 Explain the geometric meaning of the average rate of change of f over $[a, b]$ in terms of the secant line through $(a, f(a))$ and $(b, f(b))$.

11 For $f(x) = x^3$, find the average rate of change over $[-1, 2]$, and write the equation of the secant line through the corresponding points.

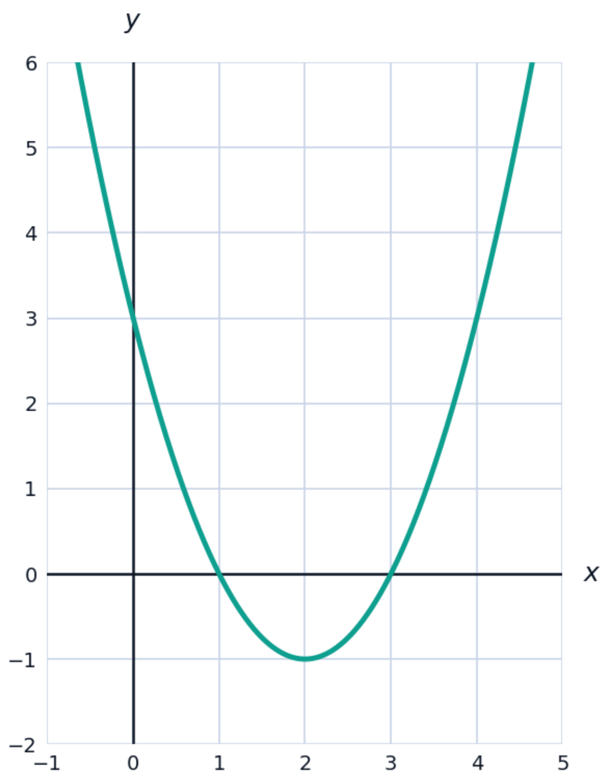
Comparing average rate of change across different functions

12 Find the average rate of change of $f(x) = 2^x$ and $g(x) = x^2$ over $[0, 4]$, and determine which function grew faster on average.

13 Find the average rate of change of $f(x) = 2^x$ over $[4, 8]$ and compare it to its value over $[0, 4]$ (found above). What does this reveal about exponential growth?

Average rate of change from a graph

14 Using the graph of $f(x) = x^2 - 4x + 3$ shown, estimate the average rate of change over $[0, 4]$ by reading endpoint values from the graph, then confirm algebraically.



Units and interpretation in context

15 A tank's water volume is $V(t) = 200 - 3t^2$ liters, where t is minutes. Find the average rate of change over $[0, 5]$, including correct units, and interpret its sign.

16 A population model gives $P(t) = 500e^{0.1t}$ (people, years). Find the average rate of change over $[0, 10]$ and state its units, rounding to the nearest whole number.